

Coherent Drawdown-Adjusted Performance: Replication and Extension Across International Equity Markets

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Abstract

Mechanical drawdown rules are among the most widely used risk management tools in institutional trading, yet their statistical properties remain poorly understood. This paper extends Varma [2025] by replicating and testing the Coherent Drawdown-Adjusted Performance (CDAP) metric out-of-sample across European equity ETFs and a range of drawdown thresholds motivated by pod shop practice. Using rolling 22-day windows over 1993–2024, we confirm that traditional Drawdown-Adjusted Performance (DAP) ranked the worst 22-day window across four major crises — the Dot-com crash, the GFC, COVID, and the 2022 rate shock — between the 8th and 25th percentile for SPY in every case, treating each event as unremarkable, while CDAP correctly ranked all four windows in the bottom 2% of the full sample. This pathology generalises: DAP ranked the same period at approximately the 32nd percentile for both EWU and EWG, while CDAP ranked it below the 0.1th percentile in both cases. Threshold sensitivity analysis shows that Millennium-style 5% drawdown rules applied to SPY since 1993 produce a terminal portfolio value of $5\times$ versus $24\times$ for buy-and-hold, with 280 individual transactions. These results suggest that mechanical drawdown rules substantially underperform buy-and-hold

benchmarks across equity markets and that CDAP offers a more coherent basis for risk-adjusted evaluation.

Keywords: drawdown rules, risk management, CDAP, pod shops, multi-strategy funds, ETFs, performance measurement

JEL Classification: G11, G23, G32

The author thanks Samir Varma for making the methodology publicly available. All errors are my own. Code and data are available at <https://github.com/tanmaya-lodhia/CDAP-Replication-and-Extension-Across-International-Equity-Markets>.

Contents

1	Introduction	2
2	Background	3
2.1	Pod Shops and Drawdown Rules in Practice	3
2.2	Drawdown-Adjusted Performance: Definition and Flaw	3
3	Methodology	4
3.1	The CDAP Metric	4
3.2	Data	5
3.3	Rolling Window Design	5
3.4	Threshold Sensitivity Backtest	6
4	Results I: Replication	6
5	Results II: European Extension	8
6	Results III: Threshold Sensitivity	10
7	Discussion	11
7.1	Why DAP Persists Despite Its Flaws	11
7.2	Implications for Pod Shop Risk Management	12
7.3	Limitations	13
8	Conclusion	13
A	Appendix A: Additional Figures	15
B	Appendix B: Window Length Robustness	17

1 Introduction

Risk management in multi-strategy hedge funds relies heavily on mechanical drawdown rules. A portfolio manager whose book falls by a fixed percentage sees capital cut or withdrawn, regardless of market context, recovery prospects, or the statistical properties of the underlying strategy. These rules are simple, transparent, and near-universal in pod shop structures such as Millennium Management, Citadel, and Point72. Their simplicity is precisely their appeal: in a firm running hundreds of semi-autonomous strategies simultaneously, a mechanical trigger removes the need for centralised judgment on every individual position.

Varma [2025] challenges the statistical foundation of this practice. Using nearly three decades of ETF data, he shows that the traditional Drawdown-Adjusted Performance (DAP) metric used to evaluate and trigger these rules is mathematically incoherent: when returns are negative, larger drawdowns paradoxically improve the DAP score, causing the metric to rank catastrophic market events as unremarkable. He introduces a corrected metric, Coherent Drawdown-Adjusted Performance (CDAP), which behaves logically across all return conditions. Applied to four major market crises — the Dot-com crash, the GFC, COVID, and the 2022 rate shock — DAP ranked the worst 22-day window between the 8th and 25th percentile in every case, while CDAP ranked all four windows in the bottom 2% of the full sample.

This paper replicates and extends Varma [2025] along two dimensions. First, we test whether the DAP pathology and CDAP correction generalise beyond US equity markets by applying the same rolling window analysis to four international ETFs covering Japan, the United Kingdom, Germany, and France. Second, we conduct a threshold sensitivity analysis examining the cumulative performance of mechanical drawdown rules at five thresholds, including the 5% and 7.5% rules specifically associated with Millennium Management, across the full sample period from 1993 to 2024.

Our results confirm both extensions. The DAP pathology is present in all five markets, with COVID crash rankings between the 29th and 32nd percentile across international ETFs under DAP and at or below the 0.3th percentile under CDAP. Threshold sensitivity analysis shows that the 5% Millennium rule produced a terminal portfolio value of $4.7\times$ versus $24.0\times$ for buy-and-hold over the sample period, with underperformance monotonically increasing as thresholds tighten. These results suggest that the underperformance of mechanical drawdown rules generalises across developed equity markets in long-only ETF backtests and is robust to a wide range of threshold specifications.

The remainder of the paper is structured as follows. Section 2 provides background on pod shop risk management and the mathematical flaw in DAP. Section 3 describes the CDAP metric, data, and empirical design. Sections 4, 5, and 6 present the three sets of results. Section 7 discusses implications and limitations. Section 8 concludes.

2 Background

2.1 Pod Shops and Drawdown Rules in Practice

Multi-strategy hedge funds, commonly known as pod shops, organise trading activity into semi-autonomous teams (pods), each allocated a defined capital budget and granted discretion over their own investment decisions. Prominent examples include Millennium Management, Citadel, and Point72, which collectively manage hundreds of billions of dollars across hundreds of independent pods. Each pod typically employs a portfolio manager and a small team of analysts running strategies across equities, fixed income, commodities, and derivatives.

Risk management in this structure is centralised at the firm level. Millennium, for instance, is widely reported to operate a strict drawdown rule: a pod that loses 5% of its allocated capital sees that allocation cut by half, while a loss of 7.5% results in the pod being shut down and its capital reallocated [Rubinstein, 2023]. These thresholds serve two purposes: limiting the firm’s aggregate exposure to any single strategy, and creating strong incentives for portfolio managers to avoid losses. The rules are simple, transparent, and universally understood within the firm.

The appeal of such rules is intuitive. In a structure managing hundreds of uncorrelated strategies simultaneously, a simple mechanical trigger allows centralised risk oversight without requiring firm-level judgment about every individual pod’s positions. However, as Varma [2025] argues, the simplicity that makes these rules operationally attractive may also make them statistically destructive, a possibility this paper examines empirically.

2.2 Drawdown-Adjusted Performance: Definition and Flaw

The traditional Drawdown-Adjusted Performance (DAP) metric is defined as:

$$\text{DAP} = \frac{r}{|d|} \quad (1)$$

where r denotes the return over a given window and $|d|$ denotes the absolute value of the maximum drawdown over the same window. Although intuitive when returns are positive, equation (1) exhibits a critical flaw when $r < 0$: larger drawdowns reduce the denominator’s absolute value relative to the numerator, paradoxically *improving* the ratio during adverse periods. As Varma [2025] notes, this produces a metric that can rank catastrophic market events as unremarkable.

The practical consequence of this flaw is severe. Consider two windows with identical negative returns of -10% . If the first window has a maximum drawdown of -5% and the second has a maximum drawdown of -30% , the first window receives a DAP score of -2.0 while the second receives -0.33 . Under DAP, the second window, which experienced

six times the peak-to-trough loss, appears to be the better risk-adjusted outcome. A risk manager relying on DAP would therefore systematically underestimate the severity of periods combining large losses with large drawdowns, precisely the conditions that define a market crisis.

DAP belongs to a broader family of drawdown-adjusted performance metrics that have been proposed as alternatives to the Sharpe ratio for strategies with non-normal return distributions. The Calmar ratio [Young, 1991] divides annualised return by maximum drawdown over a fixed lookback period, typically three years. The Sterling ratio modifies this by averaging the largest drawdowns rather than using the single maximum. The Burke ratio [Burke, 1994] takes the square root of the sum of squared drawdowns as its denominator, penalising multiple drawdowns rather than just the largest. All three share a structural vulnerability with DAP: each divides a signed return by a strictly positive drawdown measure. When the numerator is negative, a larger denominator produces a less negative ratio — regardless of whether that denominator is maximum drawdown, average drawdown, or root sum of squared drawdowns. A period combining large losses with large drawdowns therefore receives a better score than one with the same losses and smaller drawdowns, precisely the incoherence CDAP is designed to correct. Whether this pathology manifests empirically in Calmar, Sterling, and Burke ratios to the same degree as in DAP is a question for future research; the present paper focuses on DAP as the metric most directly relevant to pod shop practice.

The theoretical literature on optimal drawdown control, beginning with Grossman and Zhou [1993], focuses on minimising the probability of large peak-to-trough losses rather than on measuring them accurately. The failure of DAP identified by Varma [2025] represents a distinct problem: not how to limit drawdowns, but how to measure their severity coherently once they have occurred.

3 Methodology

3.1 The CDAP Metric

Varma [2025] introduces the Coherent Drawdown-Adjusted Performance (CDAP) metric to correct the sign-dependent flaw in DAP. The metric is defined as:

$$\text{CDAP} = r \times |d|^{-\text{sgn}(r)} \quad (2)$$

where $\text{sgn}(r)$ denotes the sign function, equal to $+1$ when $r > 0$ and -1 when $r < 0$. Equation (2) simplifies to two intuitive cases:

$$\text{CDAP} = \begin{cases} r / |d| & \text{if } r > 0 \quad (\text{reward penalised by drawdown}) \\ r \times |d| & \text{if } r < 0 \quad (\text{loss amplified by drawdown}) \end{cases} \quad (3)$$

This formulation ensures that larger drawdowns always worsen the score, regardless of the sign of returns — a property Varma [2025] terms *coherence*.

The construction of CDAP follows the broader literature on risk-adjusted performance metrics, which recognises that dividing returns by a risk measure is only well-defined when the relationship between the numerator and denominator is consistent across the sign of returns. Lo [2002] makes a related point in the context of the Sharpe ratio, noting that performance metrics based on ratios can behave unexpectedly when return distributions are non-normal — a concern that applies directly to the sign-dependent pathology in DAP.

3.2 Data

This paper uses daily adjusted closing prices for five exchange-traded funds obtained from Yahoo Finance: SPY (S&P 500, United States), EWJ (MSCI Japan), EWU (MSCI United Kingdom), EWG (MSCI Germany), and EWQ (MSCI France). The sample period runs from 29 January 1993 to December 2024 for SPY, with EWJ, EWU, EWG, and EWQ each available from 18 March 1996. Daily log returns are computed as $r_t = \ln(P_t/P_{t-1})$, where P_t denotes the adjusted closing price on day t . Adjusted prices account for dividends and stock splits, ensuring that measured drawdowns reflect the true economic experience of a buy-and-hold investor.

The choice of ETFs follows Varma [2025], who uses the same instruments as representative liquid proxies for each market. While ETF returns do not perfectly replicate the intraday books of a pod shop portfolio manager, they provide a clean, publicly available benchmark for testing whether the metric pathology identified by Varma [2025] generalises across markets. The use of adjusted closing prices introduces a minor look-ahead bias in the dividend adjustment, though this is standard practice in the empirical finance literature and is unlikely to materially affect drawdown measurement.

3.3 Rolling Window Design

All metrics are computed over rolling 22-day windows, where each window contains the daily log returns for the 22 consecutive trading days ending on date t . A window length of 22 trading days corresponds to approximately one calendar month and was selected to reproduce the annualised return of approximately -99% reported by Varma [2025] for the COVID crash window of 21 February to 23 March 2020. No look-ahead bias is introduced: each window uses only data available on or before day t .

Within each window, the cumulative log return is computed as $r = \sum_{i=1}^{22} r_i$ and

annualised by multiplying by 252/22. The maximum drawdown d is defined as the largest peak-to-trough decline in cumulative value within the window, expressed as a positive number. To prevent division by zero in cases where no drawdown occurs, a floor of 0.1% is applied to $|d|$. This assumption affects fewer than 0.1% of all windows and has no material impact on the results.

Percentile ranks are computed empirically across all rolling windows in the full sample for each ETF. A window’s percentile rank under DAP or CDAP therefore reflects its position in the historical distribution of that metric for that specific market, rather than against a theoretical distribution. This non-parametric approach follows Varma [2025] and avoids assumptions about the shape of the return distribution.

3.4 Threshold Sensitivity Backtest

To evaluate the performance of mechanical drawdown rules in practice, we simulate a rule-based investor who holds a long position in each ETF and exits when the cumulative drawdown from the most recent portfolio peak breaches a fixed threshold. Upon exit, the investor holds cash and re-enters the market after a fixed cooldown period. Three cooldown lengths are tested: 5, 10, and 20 trading days. Five drawdown thresholds are tested: 5%, 7.5%, 10%, 15%, and 20%. The 5% and 7.5% thresholds are motivated directly by Millennium’s reported capital reduction and pod closure rules respectively.

Transaction costs of 10 basis points are applied to each individual transaction, covering both entries and exits. The risk-free rate during cash periods is set to zero, which is a conservative assumption that modestly understates the relative performance of drawdown strategies during periods of positive short-term rates. All positions are assumed to execute at the same-day adjusted closing price, which overstates the precision of real-world execution but is standard in ETF backtesting given the liquidity of the instruments studied.

Performance is evaluated using three metrics: terminal portfolio value normalised to $1\times$ at the start of the sample, the Sharpe ratio computed from annualised daily returns with a risk-free rate of zero, and the maximum drawdown over the full sample period. Results are compared against a buy-and-hold benchmark for each ETF. The backtest is implemented without look-ahead bias: the drawdown trigger at each point in time is computed using only price data available on that day.

4 Results I: Replication

Table 2 and Figure 2 present the core replication results for SPY. Across 8,015 rolling 22-day windows from January 1993 to December 2024, the COVID crash window of 21 February to 23 March 2020 produced an annualised return of -99.1% and a maximum

drawdown of 33.4%. Under DAP, this window ranked at the 24.9th percentile of all historical observations, placing it in the unremarkable lower quarter of the distribution. A risk manager monitoring DAP in real time would have received no signal distinguishing this period from a routine below-average month.

Under CDAP, the same window ranked at the 0.0th percentile, placing it at the extreme left tail of the historical distribution with no prior window scoring worse. This result directly replicates the central finding of Varma [2025]: the incoherence of DAP is not a theoretical curiosity but a practically significant failure that manifests during the precise market conditions when accurate risk measurement matters most. The contrast between the two panels of Figure 2 illustrates this visually. The COVID window sits unremarkably within the body of the DAP distribution, while under CDAP it is isolated at the far left tail with a clear gap separating it from the next worst observation.

Table 1 extends the replication to four distinct stress periods in the SPY sample. Across all four crises, DAP ranks the worst 22-day window between the 8th and 25th percentile, treating each event as unremarkable relative to the full historical distribution. CDAP ranks all four windows in the bottom 2% of observations, correctly identifying each as a tail event.

Three of the four crises produce CDAP ranks below 0.6%, consistent with the acute nature of the Dot-com, GFC, and COVID crashes. The 2022 rate shock produces a somewhat higher CDAP rank of 1.51%, reflecting the different character of that episode: rather than an acute market shock, 2022 featured a prolonged grinding decline driven by monetary tightening. Any single 22-day window captures only a fraction of the cumulative stress, producing a less extreme single-window score despite the severity of the full-year drawdown.¹ Across all four episodes, DAP provides no consistent signal distinguishing crisis periods from ordinary below-average months.

Table 1: Worst 22-day window for SPY across four crisis periods. Percentile ranks computed over the full 1993–2024 sample of 8,015 windows.

Crisis	Window	Ann. Return (%)	Max DD (%)	DAP Rank (%)	CDAP Rank (%)
Dot-com	Jun–Jul 2002	−93.0	20.7	17.5	0.51
GFC	Sep–Oct 2008	−98.5	30.5	23.6	0.04
COVID	Feb–Mar 2020	−99.1	33.4	24.9	0.00
2022 rate shock	Aug–Sep 2022	−79.8	13.0	8.4	1.51

Note: Window dates refer to the worst-scoring 22-day rolling window within each crisis period. DAP ranks the worst window between the 8th and 25th percentile across all four crises, while CDAP ranks all four windows in the bottom 2% of the full sample.

¹The worst Dot-com window falls in July 2002 rather than at the index peak in March 2000, reflecting the protracted nature of the crash and the market’s final leg lower in summer 2002.

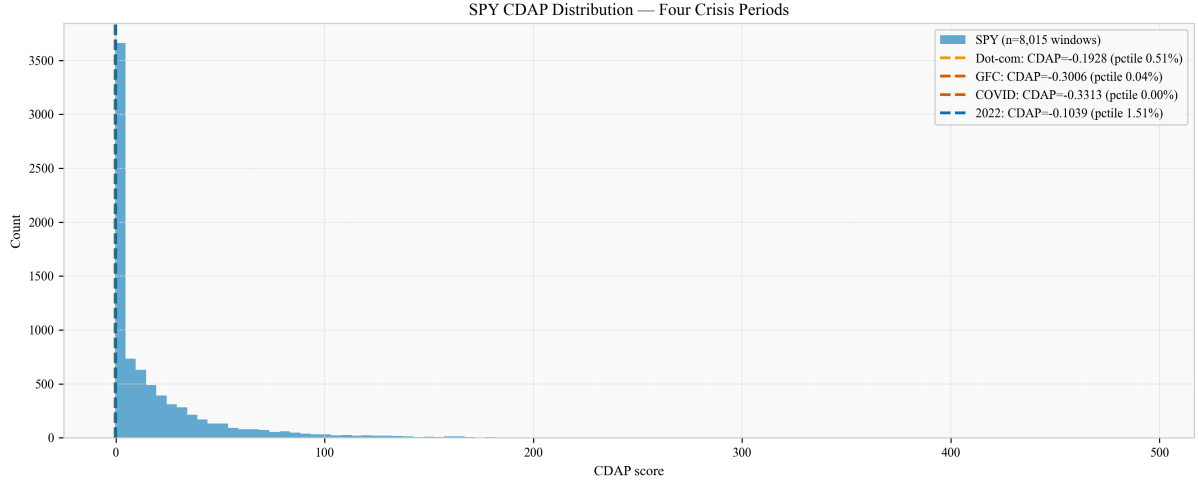


Figure 1: SPY CDAP distribution (all 8,015 rolling 22-day windows, 1993–2024) with the worst window for each of four crisis periods marked. All four crisis windows fall in the extreme left tail of the distribution.

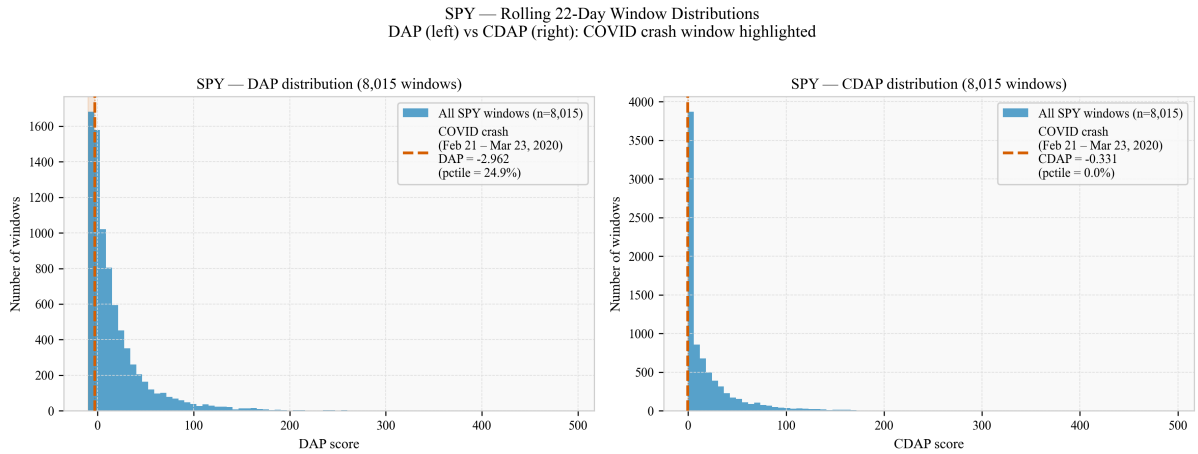


Figure 2: Rolling 22-day window distributions of DAP (left) and CDAP (right) for SPY over 8,015 windows (1993–2024). The dashed vertical line marks the COVID crash window (21 February – 23 March 2020). Under DAP, this window ranks at the 24.9th percentile; under CDAP, it ranks at the 0.0th percentile.

5 Results II: European Extension

Table 2 reports DAP and CDAP percentile ranks for the COVID crash window across all five ETFs. The DAP pathology generalises uniformly across markets: DAP ranked the COVID window between the 29th and 32nd percentile for all four international ETFs, treating the same crash as unremarkable in Japan, the United Kingdom, Germany, and France as it did in the United States. CDAP ranked the same window at or below the 0.3th percentile in all cases, correctly identifying it as a tail event regardless of market.

Two results in Table 2 are worth noting specifically. First, EWU ranked the COVID

window at exactly the 0.00th percentile under CDAP, matching SPY despite Varma [2025] identifying the UK market as one that shows no clear pattern in its drawdown probability structure. This suggests the CDAP correction operates independently of the underlying return distribution's tail properties. Second, the maximum drawdown during the COVID window was considerably larger for European ETFs (37.9% to 40.0%) than for SPY (33.4%), reflecting the additional currency and liquidity stress in European markets during the crash. Despite this difference in drawdown severity, DAP ranked all markets similarly, further illustrating its insensitivity to the information contained in drawdown magnitude.

Figure 3 plots the CDAP distribution for each international ETF with the COVID window highlighted. In all cases the pattern is consistent with the SPY replication: the COVID window sits at the extreme left of the distribution, isolated from the remainder of the sample. These results confirm that the metric pathology identified by Varma [2025] is not specific to US equity markets but represents a general property of DAP that applies across developed market ETFs.

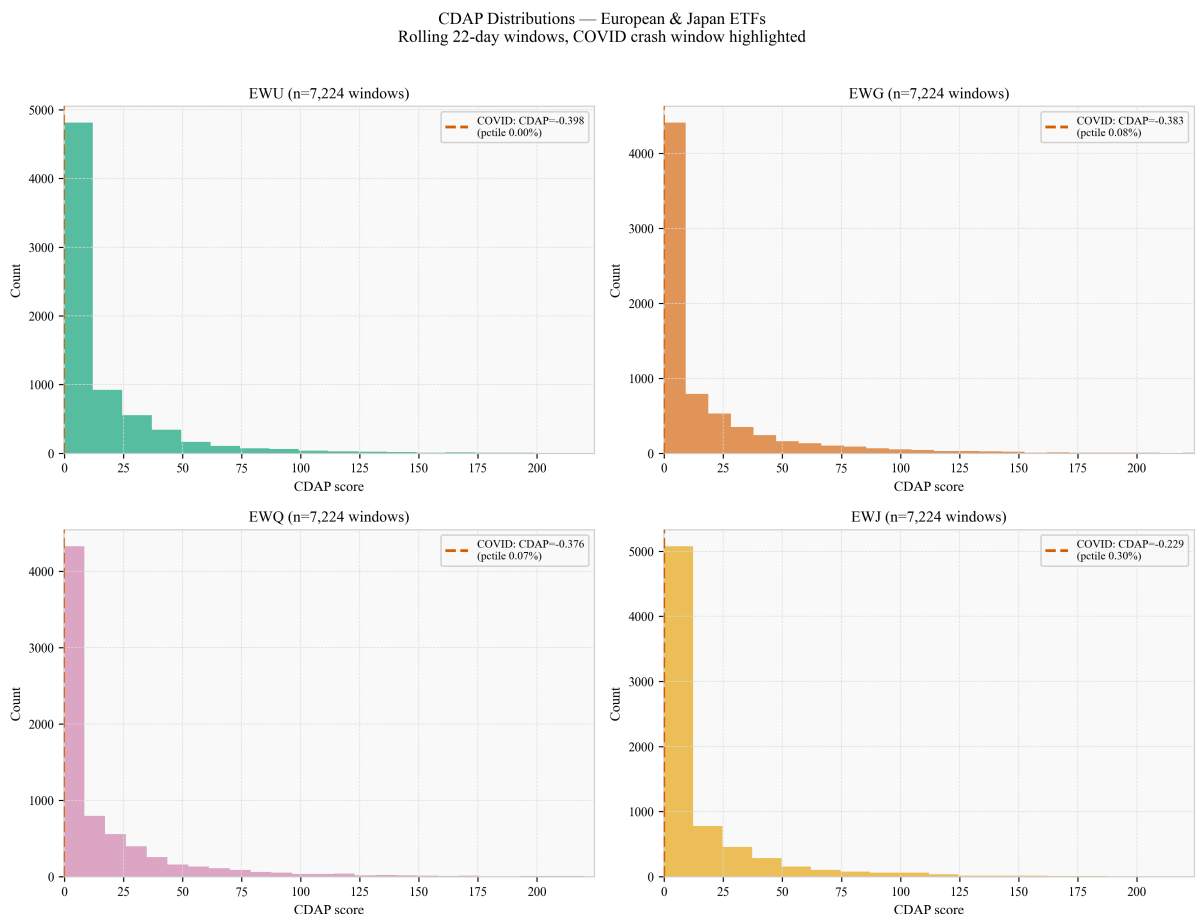


Figure 3: CDAP distributions for international ETFs (EWJ, EWU, EWG, EWQ) with the COVID crash window highlighted. In all cases the COVID window falls in the extreme left tail of the CDAP distribution, confirming that the metric generalises out-of-sample.

Table 2: COVID crash window percentile ranks by ETF and metric. Sample period 1993–2024; 22-day rolling windows.

ETF	Market	DAP Rank (%)	CDAP Rank (%)	Max DD (%)
SPY	United States	24.9	0.00	33.4
EWJ	Japan	29.1	0.30	24.3
EWU	United Kingdom	32.0	0.00	40.0
EWG	Germany	32.4	0.08	38.5
EWQ	France	31.6	0.07	37.9

Note: Max DD is the maximum drawdown within the COVID crash window. All values computed over rolling 22-day windows, 1993–2024.

6 Results III: Threshold Sensitivity

Figure 4 and Table 3 report cumulative performance for SPY under buy-and-hold and five mechanical drawdown rules from January 1993 to December 2024. The results show a monotonic relationship between threshold tightness and underperformance: every drawdown rule underperforms buy-and-hold, and tighter thresholds produce larger shortfalls.

The most striking result concerns the 5% threshold, which corresponds to Millennium’s capital reduction rule. With a 5-day cooldown, the 5% rule triggered 280 individual transactions over the sample period and produced a terminal portfolio value of $4.7\times$ the initial investment, compared to $24.0\times$ for buy-and-hold. This represents a loss of approximately 80% of the wealth that a passive investor would have accumulated. The mechanism is the one described by Varma [2025] as death by a thousand cuts: frequent exits at small losses followed by re-entries that miss partial recoveries compound into a severe cumulative shortfall over time. The Sharpe ratio of 0.39 under the 5% rule compares to 0.63 for buy-and-hold, confirming that the underperformance is not explained by reduced volatility.

The 7.5% threshold, corresponding to Millennium’s pod closure rule, produced a terminal value of $8.0\times$ with 146 transactions and a Sharpe ratio of 0.48. While less severe than the 5% rule, this still represents a substantial underperformance relative to buy-and-hold. Performance improves as the threshold widens: the 10% and 15% rules produced terminal values of $15.3\times$ and $15.8\times$ respectively, and the 20% rule came closest to matching buy-and-hold at $22.0\times$ with only 18 transactions. This pattern is consistent with the interpretation that tighter rules generate more whipsaw trades, each of which locks in a loss that a subsequent recovery would have erased. Wider rules trigger less frequently, allowing more recoveries to play out before exit.

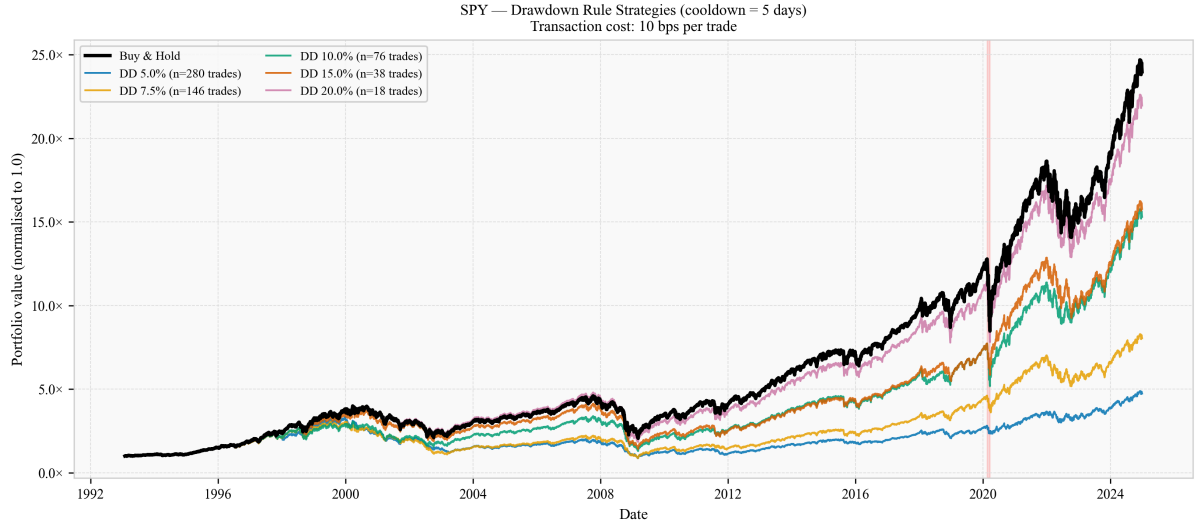


Figure 4: Cumulative portfolio value (normalised to 1.0) for SPY under buy-and-hold and mechanical drawdown rules at thresholds of 5%, 7.5%, 10%, 15%, and 20%, with a 5-day re-entry cooldown and 10 basis points transaction cost per trade. The pink vertical line marks the COVID crash. All drawdown strategies underperform buy-and-hold, with underperformance monotonically increasing as the threshold tightens.

Table 3: Threshold sensitivity summary for SPY (cooldown = 5 days, transaction cost = 10 bps). Sample period 1993–2024.

Strategy	Threshold	Trades	Terminal Value	Max DD (%)	Sharpe
Buy & Hold	—	—	24×	55.2	0.63
DD Rule	5%	280	5×	73.6	0.39
DD Rule	7.5%	146	8×	71.3	0.48
DD Rule	10%	76	15×	61.6	0.58
DD Rule	15%	38	16×	65.5	0.58
DD Rule	20%	18	22×	61.6	0.63

Note: Terminal value normalised to 1× at start of sample. Transaction costs of 10 bps applied per individual transaction. Sharpe ratio computed using daily returns, annualised, with risk-free rate set to zero.

7 Discussion

7.1 Why DAP Persists Despite Its Flaws

The persistence of DAP in institutional practice despite its mathematical incoherence is not difficult to explain. The metric is simple to compute, easy to communicate to non-technical stakeholders, and produces intuitively reasonable rankings during normal market conditions when returns are predominantly positive. For the majority of rolling windows

in any sample, DAP and CDAP will produce similar orderings, since the sign-dependent flaw only manifests when returns are negative. Institutions that backtested DAP on historical data dominated by positive-return periods would have found little reason to question it.

The behavioural finance literature offers an additional explanation. Tversky and Kahneman [1992] document that individuals exhibit loss aversion and a preference for rules that provide a sense of control during uncertain periods. A mechanical drawdown rule satisfies this preference regardless of its statistical validity: it gives portfolio managers and risk officers a defined course of action when positions move against them, reducing the psychological discomfort of holding a losing position. As Varma [2025] notes, traders often prefer the illusion of risk management over actual risk management. Replacing DAP with CDAP addresses the measurement problem but does not resolve the underlying behavioural demand for simple, actionable rules.

7.2 Implications for Pod Shop Risk Management

The results of this paper present a tension for pod shop risk managers. On one hand, the threshold sensitivity analysis provides clear empirical evidence that the specific rules used by Millennium substantially underperform passive investment in the underlying market in long-only ETF backtests. On the other hand, the purpose of these rules within a pod shop is not solely to maximise the performance of any individual pod. Centralised drawdown limits also serve to contain the firm’s aggregate tail risk, prevent correlated losses across pods during market stress, and align the incentives of portfolio managers with those of the firm’s investors. The results here do not resolve that tension, but they do suggest the cost of mechanical rules is higher than practitioners may appreciate.

For portfolio managers operating within pod shop structures, the practical implication is that CDAP offers a more informative basis for self-evaluation than DAP. A PM whose strategy experiences a large drawdown during a period of broad market stress should be evaluated differently from one whose drawdown is idiosyncratic. CDAP captures this distinction directly: a large drawdown accompanied by a large negative return produces a worse CDAP score than the same drawdown in a period of flat returns, whereas DAP treats both identically. PMs who can demonstrate strong CDAP-adjusted performance during stress periods have a more defensible case against mechanical capital cuts than those relying on DAP.

For allocators evaluating multi-strategy funds, the DAP pathology documented here has a direct implication for manager due diligence. Risk reports that rely on DAP or structurally similar metrics will systematically understate the severity of drawdowns during market crises. Allocators should request CDAP-adjusted performance attribution, particularly for periods of broad market stress, before drawing conclusions about a fund’s risk-adjusted track record.

More broadly, Varma [2025] proposes three principles for a context-sensitive framework: incorporating volatility regime information, using cross-asset confirmation before triggering exits, and applying dynamic percentile-based thresholds rather than fixed cutoffs. The CDAP metric provides a more coherent basis for the risk measurement underlying such a framework. Whether pod shops adopt more sophisticated approaches in practice remains an open question, as the operational simplicity of mechanical rules carries its own institutional value.

7.3 Limitations

Several limitations of this analysis should be noted. First, ETF returns are used as proxies for pod shop trading books. In practice, pod portfolios are typically long-short, leveraged, and concentrated in individual securities rather than diversified index products. The results therefore speak to the properties of drawdown rules applied to liquid index ETFs, which may differ from their properties in the context of individual stock or derivatives portfolios.

Second, the 22-day window length is an assumption not explicitly stated by Varma [2025]. While this choice successfully reproduces his reported annualised return for the COVID crash window, Appendix B reports results for 10-day and 44-day windows across all five ETFs. The CDAP finding is stable across all 15 combinations, with no rank exceeding the 0.39th percentile, confirming that the core result does not depend on the specific window length chosen. Third, trade execution is assumed to occur at the same-day adjusted closing price, which overstates the precision of real-world exits during volatile periods. Fourth, the risk-free rate during cash periods is set to zero throughout the sample, which modestly understates the relative performance of drawdown strategies during periods of elevated short-term rates such as 2022 to 2024. Fifth, the backtest covers a single asset class in developed markets only. Whether the results generalise to emerging markets, fixed income, or commodity ETFs is a question for future research.

8 Conclusion

This paper replicates and extends Varma [2025] across international equity markets and a range of drawdown thresholds motivated by pod shop practice. Using rolling 22-day windows over 1993 to 2024, we confirm that traditional DAP ranked the COVID crash of February to March 2020 at the 24.9th percentile for SPY, while CDAP correctly ranked it at the 0.0th percentile. This finding generalises uniformly to European and Japanese equity ETFs, where DAP ranked the same crash between the 29th and 32nd percentile and CDAP ranked it at or below the 0.3th percentile in all cases.

Threshold sensitivity analysis shows that Millennium-style drawdown rules substantially

underperform buy-and-hold over a three-decade horizon in long-only ETF backtests. The 5% rule produced a terminal portfolio value of $4.7\times$ versus $24.0\times$ for buy-and-hold, with 280 individual transactions and a Sharpe ratio of 0.39 compared to 0.63. Underperformance is monotonically increasing as thresholds tighten, consistent with the whipsaw mechanism described by Varma [2025].

These results suggest that the underperformance of mechanical drawdown rules extends beyond US equity markets in long-only ETF backtests and is robust to a range of threshold specifications. The CDAP metric offers a more coherent basis for risk-adjusted performance measurement across market conditions. Future work could extend this analysis to intraday data, individual stock portfolios, or alternative asset classes, and could test whether dynamic threshold frameworks of the kind proposed by Varma [2025] deliver measurable improvements over static rules in live trading environments.

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A Appendix A: Additional Figures

Figures 5–8 present supplementary results supporting the main analysis.

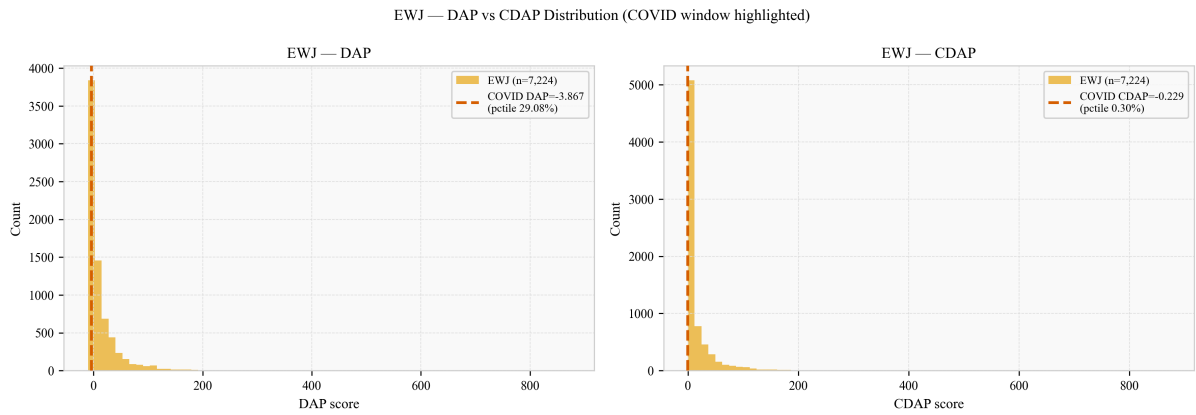


Figure 5: DAP and CDAP distributions for EWJ (Japan) with the COVID crash window highlighted. DAP ranks the window at the 29.1th percentile; CDAP ranks it at the 0.30th percentile.

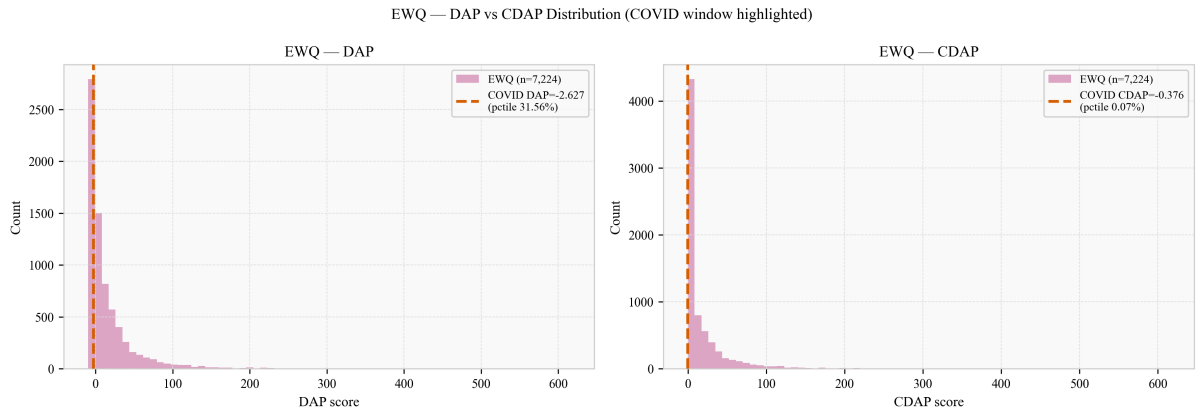


Figure 6: DAP and CDAP distributions for EWQ (France) with the COVID crash window highlighted. DAP ranks the window at the 31.6th percentile; CDAP ranks it at the 0.07th percentile.

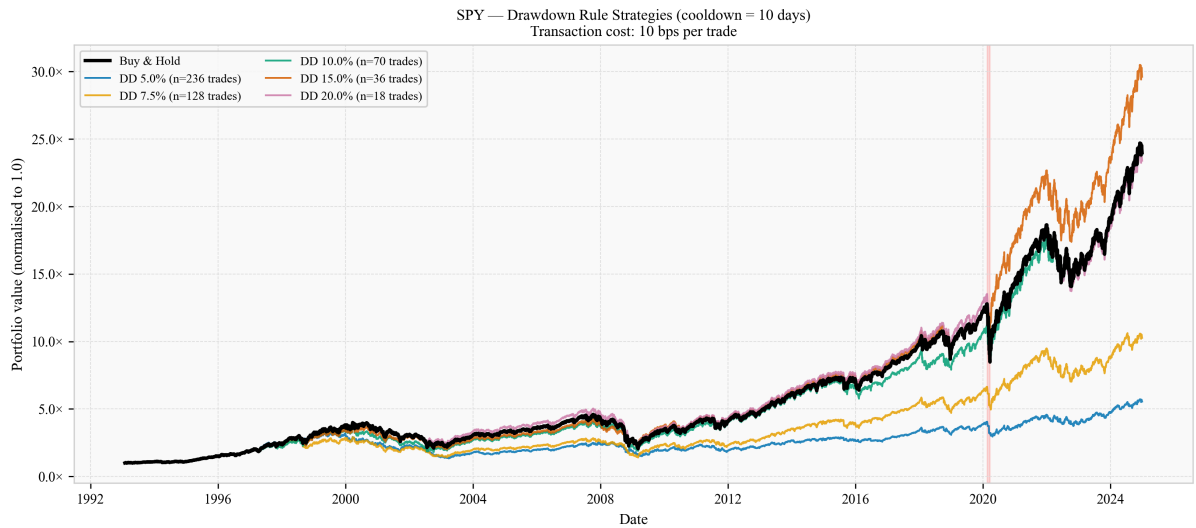


Figure 7: Cumulative portfolio value for SPY under buy-and-hold and mechanical draw-down rules with a 10-day re-entry cooldown and 10 basis points transaction cost per trade.

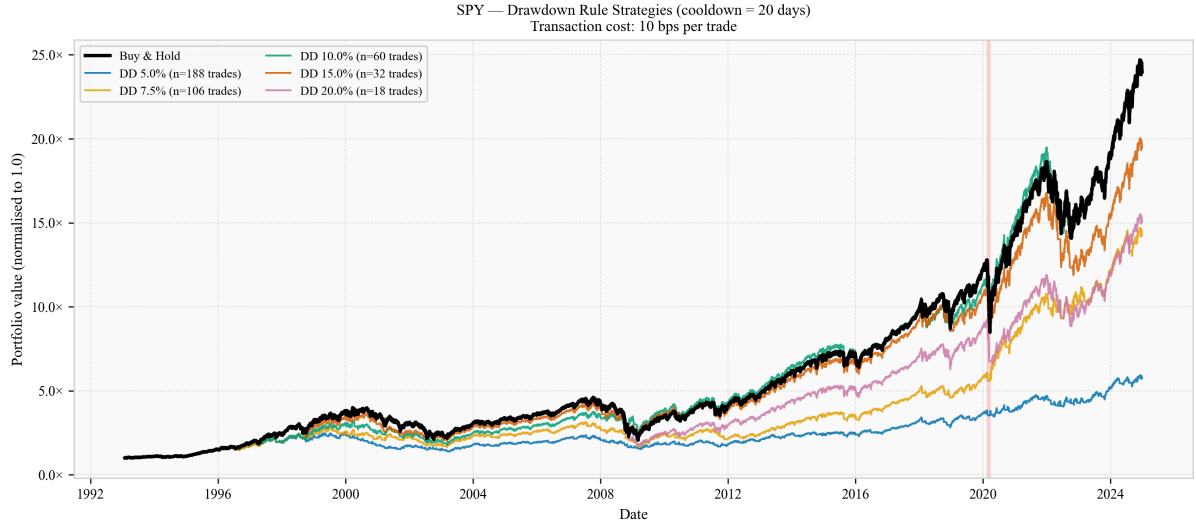


Figure 8: Cumulative portfolio value for SPY under buy-and-hold and mechanical draw-down rules with a 20-day re-entry cooldown and 10 basis points transaction cost per trade.

B Appendix B: Window Length Robustness

Table 4 reports DAP and CDAP percentile ranks for the COVID crash window across all five ETFs and three rolling window lengths: 10, 22, and 44 trading days. The core finding is robust across all 15 combinations: CDAP ranks the COVID window between the 0.00th and 0.39th percentile in every case, confirming that the extreme tail classification is not an artefact of the 22-day window assumption.

Two patterns in Table 4 are worth noting. First, DAP ranks improve at longer windows: SPY’s DAP rank falls from 31.5% at 10 days to 24.9% at 22 days to 11.1% at 44 days. Longer windows smooth the crash into a larger trend move, reducing the severity of the negative return relative to the drawdown and paradoxically making DAP appear more sensitive to the event. This is a further illustration of the metric’s incoherence rather than evidence against the critique. Second, CDAP ranks are stable across all window lengths and markets, ranging from 0.00% to 0.39%, confirming that the correction behaves consistently regardless of the measurement horizon chosen.

Table 4: Window length robustness: COVID crash window percentile ranks across rolling window lengths of 10, 22, and 44 trading days.

ETF	Window	DAP Rank (%)	CDAP Rank (%)	Ann. Return (%)	Max DD (%)
SPY	10-day	31.5	0.10	−99.4	22.2
SPY	22-day	24.9	0.00	−99.1	33.4
SPY	44-day	11.1	0.10	−89.4	33.7
EWJ	10-day	36.4	0.28	−94.0	15.9
EWJ	22-day	29.1	0.30	−94.1	24.3
EWJ	44-day	17.2	0.26	−81.2	28.0
EWU	10-day	37.9	0.07	−99.9	27.7
EWU	22-day	32.0	0.00	−99.7	40.0
EWU	44-day	18.8	0.03	−95.7	42.3
EWG	10-day	37.9	0.12	−99.8	26.5
EWG	22-day	32.4	0.08	−99.5	38.5
EWG	44-day	19.7	0.39	−93.4	39.6
EWQ	10-day	37.1	0.12	−99.7	25.3
EWQ	22-day	31.6	0.07	−99.5	37.9
EWQ	44-day	17.5	0.22	−93.5	39.2

Note: 22-day results replicate Table 2. All other rows use identical methodology with the window length as stated. No CDAP rank exceeds 1.0% across all 15 combinations.

SPY CDAP Distribution — Window Length Sensitivity
 COVID crash window (21 Feb – 23 Mar 2020) highlighted in each panel

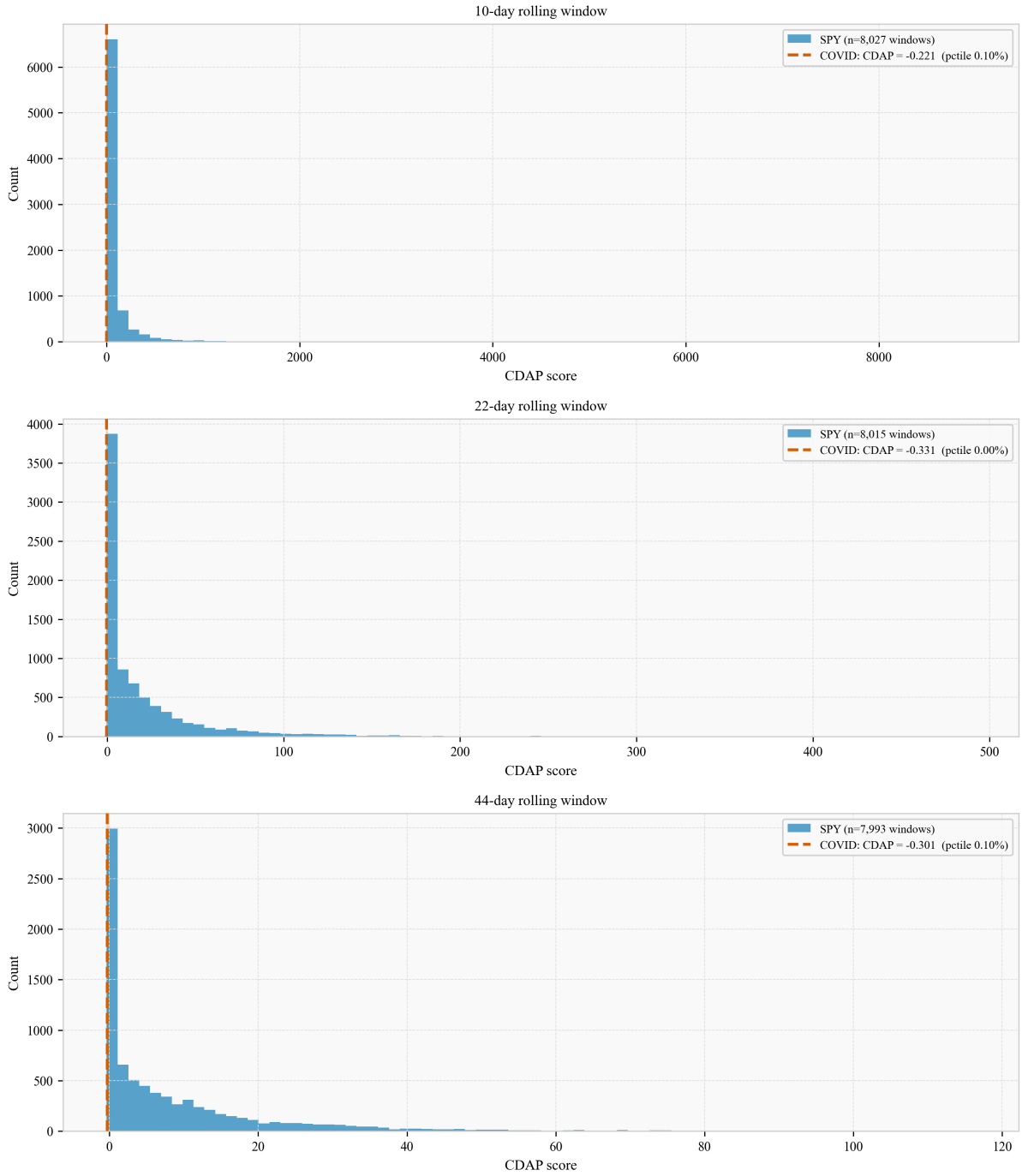


Figure 9: SPY CDAP distributions for rolling window lengths of 10, 22, and 44 trading days, with the COVID crash window highlighted in each panel. The COVID window remains in the extreme left tail across all three specifications.